Worksheet 3.3 — Derivatives of Logarithmic Functions

Grade 12 Calculus and Vectors (MCV4U) · Unit 3: Derivatives of Transcendental Functions

The Big Ideas

$\frac{d}{dx}\ln x = \frac{1}{x}$; $\frac{d}{dx}\ln(g) = \frac{g'}{g}$; $\frac{d}{dx}\log_b x = \frac{1}{x\ln b}$; logarithmic differentiation for x^x etc.

- $\frac{d}{dx}\ln x = \frac{1}{x}$.
- $\frac{d}{dx}\ln(g(x)) = \frac{g'(x)}{g(x)}$.
- Logarithmic differentiation: take ln, differentiate implicitly, solve.

The Big Picture

The natural log

$\frac{d}{dx}\ln x = \frac{1}{x}$ the derivative that fills the gap the power rule leaves at x^{-1}.

General logs

$\frac{d}{dx}\log_b x = \frac{1}{x\ln b}$, with the factor $\ln b$ coming from the change of base.

With the chain rule

$\frac{d}{dx}\ln(g(x)) = \frac{g'(x)}{g(x)}$, dividing by the inside function.

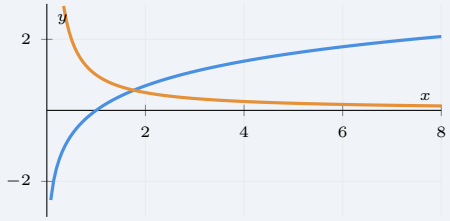
Worked Examples

Example 1 — Natural log

Differentiate $f(x) = \ln x$.

Solution. **Step 1 (recall the basic log derivative):** $\frac{d}{dx}\ln x = \frac{1}{x}$.

Conclusion: $f'(x) = \frac{1}{x}$ — the derivative of $\ln x$ is the curve $\frac{1}{x}$:



Example 2 — Chain rule

Differentiate $f(x) = \ln(x^2 + 1)$.

Solution. **Step 1 (identify outer and inner):** outer is $\ln(\,\cdot\,)$, inner is $g(x) = x^2 + 1$ (so $g'(x) = 2x$).

Step 2 (apply $\frac{d}{dx}\ln(g) = \frac{g'}{g}$): $\dfrac{2x}{x^2+1}$.

Conclusion: $f'(x) = \dfrac{2x}{x^2+1}$.

Example 3 — Product rule

Differentiate $f(x) = x \ln x$.

Solution. **Step 1 (identify the two factors):** $u = x$ (so $u' = 1$) and $v = \ln x$ (so $v' = \frac{1}{x}$).

Step 2 (apply the product rule $u'v + uv'$): $(1)\ln x + x \cdot \dfrac{1}{x}$.

Step 3 (simplify $x \cdot \frac{1}{x} = 1$): $\ln x + 1$.

Conclusion: $f'(x) = \ln x + 1$.

Example 4 — Other base

Differentiate $f(x) = \log_2 x$.

Solution. **Step 1 (recall the rule for a general base):** $\dfrac{d}{dx}\log_b x = \dfrac{1}{x \ln b}$; here $b = 2$.

Conclusion: $f'(x) = \dfrac{1}{x \ln 2}$.

Example 5 — Logarithmic differentiation

Differentiate $y = x^x$.

Solution. **Step 1 (take $\ln$ of both sides, since the variable is in the exponent):** $\ln y = \ln(x^x) = x \ln x$.

Step 2 (differentiate both sides implicitly; the right side needs the product rule): $\dfrac{y'}{y} = (1)\ln x + x \cdot \dfrac{1}{x} = \ln x + 1$.

Step 3 (isolate y' by multiplying both sides by y): $y' = y(\ln x + 1)$.

Step 4 (substitute back $y = x^x$): $y' = x^x(\ln x + 1)$.

Conclusion: $y' = x^x(\ln x + 1)$.

Example 6 — Chain rule

Differentiate $f(x) = \ln(3x)$.

Solution. **Step 1 (identify outer and inner):** outer is $\ln(\,\cdot\,)$, inner is $g(x) = 3x$ (so $g'(x) = 3$).

Step 2 (apply $\frac{g'}{g}$): $\dfrac{3}{3x}$.

Step 3 (simplify): $\dfrac{3}{3x} = \dfrac{1}{x}$.

Conclusion: $f'(x) = \dfrac{1}{x}$.

Example 7 — Chain / log law

Differentiate $f(x) = \ln(x^3)$.

Solution. **Step 1 (identify outer and inner):** inner is $g(x) = x^3$ (so $g'(x) = 3x^2$).

Step 2 (apply $\frac{g'}{g}$): $\dfrac{3x^2}{x^3}$.

Step 3 (simplify): $\dfrac{3x^2}{x^3} = \dfrac{3}{x}$.

Conclusion: $f'(x) = \dfrac{3}{x}$ (matches the shortcut: $\ln(x^3) = 3\ln x \Rightarrow \frac{3}{x}$).

Example 8 — Product rule

Differentiate $f(x) = x^2 \ln x$.

Solution. **Step 1 (identify the two factors):** $u = x^2$ (so $u' = 2x$) and $v = \ln x$ (so $v' = \frac{1}{x}$).

Step 2 (apply the product rule): $2x \ln x + x^2 \cdot \dfrac{1}{x}$.

Step 3 (simplify $x^2 \cdot \frac{1}{x} = x$): $2x \ln x + x$.

Conclusion: $f'(x) = 2x \ln x + x$.

Example 9 — Chain rule

Differentiate $f(x) = \ln(\cos x)$.

Solution. **Step 1 (identify outer and inner):** outer is $\ln(\,\cdot\,)$, inner is $g(x) = \cos x$ (so $g'(x) = -\sin x$).

Step 2 (apply $\frac{g'}{g}$): $\dfrac{-\sin x}{\cos x}$.

Step 3 (recognize the ratio as $\tan x$): $\dfrac{\sin x}{\cos x} = \tan x$.

Conclusion: $f'(x) = -\tan x$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = \ln(3x)$.

Question 2

Differentiate $f(x) = \ln(x^3)$.

Question 3

Differentiate $f(x) = x^2 \ln x$.

Question 4

Differentiate $f(x) = \ln(\cos x)$.

Question 5

Slope of $y = \ln x$ at $x = 2$?

Question 6

Differentiate $f(x) = \ln(x^2 + 4)$.

Question 7

Differentiate $f(x) = \log_3 x$.

Question 8

Differentiate $f(x) = \ln(5x - 1)$.

Question 9

Differentiate $f(x) = \dfrac{\ln x}{x}$ (quotient rule).

Question 10

Differentiate $f(x) = x^3 \ln x$.

Question 11

Differentiate $f(x) = \ln(e^x)$ (simplify first).

Question 12

Slope of $y = \ln x$ at $x = 1$?

Question 13 — Challenge

Differentiate $y = x^{2x}$ by logarithmic differentiation.

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = \ln(3x)$.
2. Differentiate $f(x) = \ln(x^3)$.
3. Differentiate $f(x) = x^2 \ln x$.
4. Differentiate $f(x) = \ln(\cos x)$.
5. Slope of $y = \ln x$ at $x = 2$?
6. Differentiate $f(x) = \ln(x^2 + 4)$.
7. Differentiate $f(x) = \log_3 x$.
8. Differentiate $f(x) = \ln(5x - 1)$.
9. Differentiate $f(x) = \dfrac{\ln x}{x}$ (quotient rule).
10. Differentiate $f(x) = x^3 \ln x$.
11. Differentiate $f(x) = \ln(e^x)$ (simplify first).
12. Slope of $y = \ln x$ at $x = 1$?
13. Differentiate $y = x^{2x}$ by logarithmic differentiation.

✓ Answer Key

1. $\dfrac{1}{x}$
2. $\dfrac{3}{x}$
3. $2x \ln x + x$
4. $-\tan x$
5. $\dfrac{1}{2}$
6. $\dfrac{2x}{x^2 + 4}$
7. $\dfrac{1}{x \ln 3}$
8. $\dfrac{5}{5x - 1}$
9. $\dfrac{1 - \ln x}{x^2}$
10. $3x^2 \ln x + x^2$
11. 1 (since $\ln(e^x) = x$)
12. 1
13. $x^{2x}(2 \ln x + 2)$